

Impact Ionization Coefficients in 4H-SiC

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Abstract—Photomultiplication measurements using 244- and 325-nm excitation have been undertaken on a series of thick 4H-SiC avalanche diodes. With avalanche widths of between 2.7 and 6 μm and the ability to measure multiplication as low as 1.001, a much wider electric field range has been covered than reported to date. The results show that the hole ionization coefficient (β) can be obtained with a high degree of accuracy down to electric fields as low as ~ 0.9 MV/cm. The value of electron ionization coefficient (α) has been determined from mixed carrier multiplication characteristics, and the β/α ratio is found to increase significantly with decreasing electric fields. Ionization coefficients are parameterized over the electric field range from 0.9 to 5 MV/cm, enabling the multiplication and breakdown characteristics of 4H-SiC to be predicted accurately.

Index Terms—Avalanche multiplication, avalanche photodiodes (APDs), breakdown voltage, impact ionization, ionization coefficients, local model.

I. INTRODUCTION

SILICON CARBIDE's (SiC) exceptionally high breakdown field, high carrier saturation velocity, and high thermal conductivity [1], [2] can enable the realization of high-temperature and high-power applications which are well beyond the limits of much more readily available conventional silicon (Si)-based electronics. SiC is also a potential alternative to the III-nitride-based avalanche photodiodes (APDs) for ultraviolet (UV) detection [3], [4], owing to its wide bandgap and widely different ionization coefficients for electrons and holes. In addition, SiC is more technologically mature and can be grown lattice-matched on SiC substrates resulting in extremely low dark currents even at elevated temperatures.

In both power devices and APD applications, an accurate knowledge of the impact ionization coefficients is critical for device design and optimization as they determine the increase in carrier density due to the impact ionization process, the breakdown voltage, and avalanche excess noise factor.

However, the published impact ionization coefficients [5]–[11] in 4H-SiC show wide variations. Konstantinov *et al.* [5], [6] reported the first impact ionization coefficients in 4H-SiC from dc mode photomultiplication measurements on three thick $n - p^+$ diodes with base doping varying from 5.2×10^{16} to $1.2 \times 10^{18} \text{ cm}^{-3}$. Raghunathan and Baliga [7], [8] used a phase-sensitive photomultiplication measurement coupled with the pulsed electron beam induced current technique to obtain β in 4H-SiC and 6H-SiC Schottky barrier diodes. However, the value of β measured at room temperature was significantly lower than those reported by Konstantinov *et al.* [5], [6]. DC mode photomultiplication measurements were performed by Hatakeyama *et al.* [10] to investigate the avalanche breakdown behavior of $p^+ - n$ diodes on (0001) and (1120) 4H-SiC. The impact ionization coefficients were then obtained by fitting to the multiplication versus voltage characteristics and breakdown-doping density curve. A theoretical study of β in 4H-SiC was performed by Bellotti *et al.* [11] using a full-band Monte Carlo simulation.

The work by Ng *et al.* [9] is the only report of impact ionization coefficients obtained from the photomultiplication measurements using both short (230-nm) and long (365-nm) wavelengths on the same devices. A nonlocal model, which accommodates the dead space effect, nonuniform electric field, and wavelength-dependent carrier injection profiles, was employed to interpret both the multiplication and excess noise characteristics in their submicrometer 4H-SiC devices. Good agreement with the experimental results was obtained by using electron and hole ionization threshold energies of 12 and 8 eV, respectively. However, the electric field range investigated was limited to 1.82–4 MV/cm due to the thin avalanche widths of 0.105 and 0.285 μm in their devices. These results may not be accurate when applied to the lower electric fields in the thicker structures of 4H-SiC power devices.

Extrapolating the values of these published ionization coefficients [5]–[11] to the lower electric fields typical of power device operation results in β that differs by up to two orders of magnitude at a field of 1.4 MV/cm. In order to resolve this issue and to more accurately determine the impact ionization coefficients over a wider electric field range, the avalanche multiplication characteristics of three 4H-SiC APDs with nominal avalanche widths of 3, 2.7, and 6 μm were studied in this paper. The multiplication characteristics of these devices under different carrier injection conditions were obtained by using both dc and phase-sensitive ac photomultiplication measurements with 244- and 325-nm UV light. A local model is employed to interpret these multiplication characteristics and to extract the low field impact ionization coefficients in these 4H-SiC bulk structures.

Manuscript received December 14, 2007; revised April 8, 2008. The review of this paper was arranged by Editor J. Zhao.

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Digital Object Identifier 10.1109/TED.2008.926679

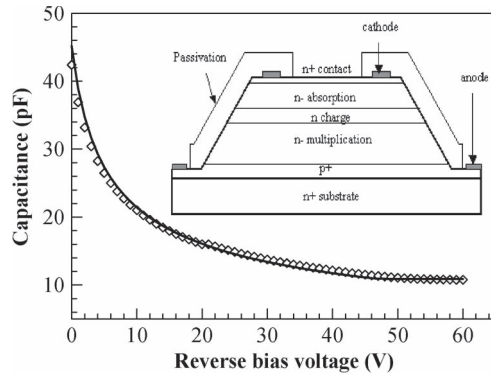


Fig. 1. Measured (symbols) and modeled (lines) capacitance-voltage characteristics of sample B with device areas of 0.25 mm^2 . The structure is shown schematically in the inset.

II. EXPERIMENT

A. Structure Details

Three different 4H-SiC APD structures used in this paper are labeled as samples A, B, and C. Sample A consists of a $0.4\text{-}\mu\text{m}$ n^+ cap layer ($N_D = 5 \times 10^{18} \text{ cm}^{-3}$), a $3\text{-}\mu\text{m}$ n^- epilayer ($N_D = 5 \times 10^{16} \text{ cm}^{-3}$), and a $1.0\text{-}\mu\text{m}$ -thick p epilayer ($N_A = 10^{17} \text{ cm}^{-3}$) grown on a p-type substrate. As shown schematically in the inset of Fig. 1, sample B was designed as a separate absorption multiplication APD (SAM-APD) which comprises a $0.2\text{-}\mu\text{m}$ n^+ contact layer ($N_D = 4 \times 10^{18} \text{ cm}^{-3}$), a $1.35\text{-}\mu\text{m}$ absorption layer ($N_D = 7.5 \times 10^{15} \text{ cm}^{-3}$), a $0.45\text{-}\mu\text{m}$ charge layer ($N_D = 4 \times 10^{17} \text{ cm}^{-3}$), a $2.7\text{-}\mu\text{m}$ multiplication layer ($N_D = 7.5 \times 10^{15} \text{ cm}^{-3}$), and a $2\text{-}\mu\text{m}$ p^+ layer ($N_A = 2 \times 10^{18} \text{ cm}^{-3}$) grown on an n^+ -doped substrate. The thickness and doping of the n-charge layer in sample B after growth were such that the device did not achieve reach-through condition (i.e., only the multiplication layer is depleted). Sample C, a reach-through SAM-APD, is made up of a $0.2\text{-}\mu\text{m}$ n^+ cap layer, a $2.0\text{-}\mu\text{m}$ absorption layer, a $0.3\text{-}\mu\text{m}$ charge layer, a $6.0\text{-}\mu\text{m}$ multiplication layer, and a $2\text{-}\mu\text{m}$ p^+ layer grown on an n^+ -doped substrate. The intended doping levels of sample C were 10^{19} , 10^{16} , 3×10^{17} , 10^{16} , and 10^{18} cm^{-3} for the n^+ , n, n, n^- , and p^+ layers, respectively.

The $1 \times 1 \text{ mm}^2$ square mesa diodes of sample A were fabricated using inductively coupled plasma etching which resulted in a negative beveled mesa with a slope angle of $\theta = 20^\circ\text{--}30^\circ$. In contrast, a $10^\circ\text{--}15^\circ$ positive bevel edge termination technology was employed in samples B and C with diode areas of 0.25 and 0.322 mm^2 , respectively. The metal contact of sample A comprises a multiple-layered stack of Ni/Ti/Mo/Au for the n-type cathode and Al/Ti/Pt/Au for the p-type anode. For samples B and C, Ni was sputtered to form the n-type cathode contacts, whereas a multiple-layered stack of Ti/Al/Ti/Ni was deposited to form the p-type anode contacts. Both contacts were annealed simultaneously at 1050°C in a N_2 ambient. Optical access windows were formed on the top ohmic contacts, and all the devices were passivated using a thermally grown SiO_2 film with an additional plasma-enhanced low-pressure chemical vapor deposition SiO_2 film on top [12].

B. Characterization Techniques

The spectral response characteristics of the devices were obtained using a Mercury-Xenon (Hg-Xe) lamp, a 0.22-m monochromator, and a lock-in amplifier. The output from the monochromator was modulated at 180 Hz with a mechanical chopper and was focused onto the optical window of the devices. A commercial UV-enhanced Si photodiode was used to calibrate the optical system in order to determine the absolute spectral response of the device under test (DUT).

The photomultiplication characteristics of the devices were investigated by illuminating the optical access window of the DUT with three sources: 1) 325-nm excitation light from a HeCd laser; 2) 244-nm light from a frequency-doubled argon ion laser; and 3) $230\text{--}325\text{-nm}$ light from the a Hg-Xe lamp. The excitation source was focused down to a diameter of $\sim 10 \text{ }\mu\text{m}$ (using the laser) or $\sim 100 \times 80 \text{ }\mu\text{m}$ (using the monochromator) in order to avoid unnecessary illumination of the mesa edge which, in turn, can give rise to mixed carrier injection or anomalous edge effects. Both dc and phase-sensitive ac measurements, which used a lock-in amplifier and modulated UV light, were performed to ensure that the multiplication characteristics were reproducible. The incident laser power was varied by inserting neutral density filters to ensure that the multiplication characteristics obtained were independent of the excitation power. At least three devices of each sample were tested to ensure the reproducibility and consistency of the results. The accuracy of multiplication data was further confirmed by scanning the small laser spot across the central region of each device and ensuring that identical values of multiplication were obtained, thereby ruling out the presence of defects of microplasmas affecting the measurement.

The multiplication factor was obtained by normalizing the measured photocurrent to the unmultiplied primary photocurrent linearly extrapolated from the measured photocurrent at low bias. This linear correction eliminates the effect of the increase in collection efficiency of the primary carriers resulting from the movement of the depletion region edge toward the top surface [13]. A 2-D raster scan was performed by focusing the 325-nm light (diameter of $\sim 5 \text{ }\mu\text{m}$) onto the DUT, and the photocurrents corresponding to each spot position were recorded at 0 and -300 V . The photocurrents extracted dropped dramatically once the laser spot was near the device edge or across the bonding pad. In general, the extracted photocurrents were uniform to within $\pm 5\%$ at both biases.

III. EXPERIMENTAL RESULTS

A. C - V and I - V Characteristics

In order to determine the layer widths and the doping levels of the DUTs, the capacitance-voltage (C - V) measurements were performed and modeled by solving Poisson's equation within the depletion approximation. An $n^+ \text{--} n \text{--} i \text{--} p \text{--} p^+$ structure, with dopings of 5×10^{18} , 6.65×10^{16} , 5.0×10^{15} (i-region), 1.18×10^{17} , and $5 \times 10^{18} \text{ cm}^{-3}$, respectively, gave the best fit to the measured C - V profile of sample A. The widths of the n, i, and p layers were estimated to be 2.8 , 0.2 , and $1.0 \text{ }\mu\text{m}$, respectively. The i-region was not part of the original design

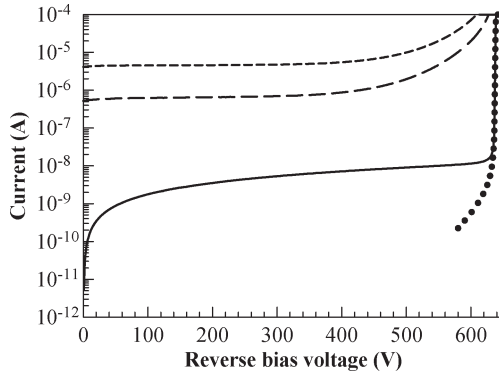


Fig. 2. (Solid line) Typical reverse dark current and photocurrents of sample B obtained using (long-dashed line) 244- and (short-dashed line) 325-nm UV light. The dotted line indicates the actual multiplied reverse dark current near breakdown.

but may have resulted from some doping compensation at the interface. The measured C - V profile of sample B, as shown in Fig. 1, is fitted with a p-i-n structure, and the doping levels are 1.5×10^{18} and $2.7 \times 10^{17} \text{ cm}^{-3}$, respectively, for p and n layers. The width and doping level in the i-region were determined to be $w = 2.7 \text{ } \mu\text{m}$ and $7.5 \times 10^{15} \text{ cm}^{-3}$, respectively. From the C - V profiles of sample B, the doping density of the n-charge layer is high enough to prevent the electric field from fully depleting it prior to breakdown. A simpler n-p structure with doping values of 1.06×10^{16} and $1.03 \times 10^{18} \text{ cm}^{-3}$ was used to fit the measured C - V profile of sample C. The extracted parameters are in good agreement with the intended values for all samples, and the diffusion voltage was estimated to be 2.9 V from extrapolating the C - V measurements, which is consistent with the reported value in [9].

Current-voltage (I - V) measurements were performed in the dark using a Keithley 237 source-measure-unit. All the diodes exhibited very low dark currents prior to avalanche breakdown, and the measured photocurrents were at least two orders of magnitude larger than the dark current as shown in Fig. 2. The relatively high dark current shown here is due to a shunt parallel resistance of $\sim 5 \times 10^{10} \text{ } \Omega$ from the photomultiplication setup used, and the true dark currents are on the order of tens of picoamperes. Also shown in Fig. 2 is the dark current near breakdown with the effect of the shunt leakage current being eliminated. The estimated breakdown voltages from dark I - V measurements were 500 and 620 V for samples A and B, respectively. Forward I - V measurements demonstrated that the photocurrent results would not be affected by series resistance over the current range investigated.

B. Room Temperature Characteristics

A maximum quantum efficiency of $\sim 59\%$ at 281 nm with a corresponding responsivity of 134 mA/W was obtained for sample A, which is close to the published result by Burr *et al.* [14], whereas an efficiency of $\sim 54\%$ at 288 nm with peak responsivity of 126 mA/W was measured for sample C. The spectral responsivities of both samples A and C drop by at least two orders of magnitude between 300 and 400 nm with cutoff at $\sim 380 \text{ nm}$, corresponding to the 3.26-eV bandgap

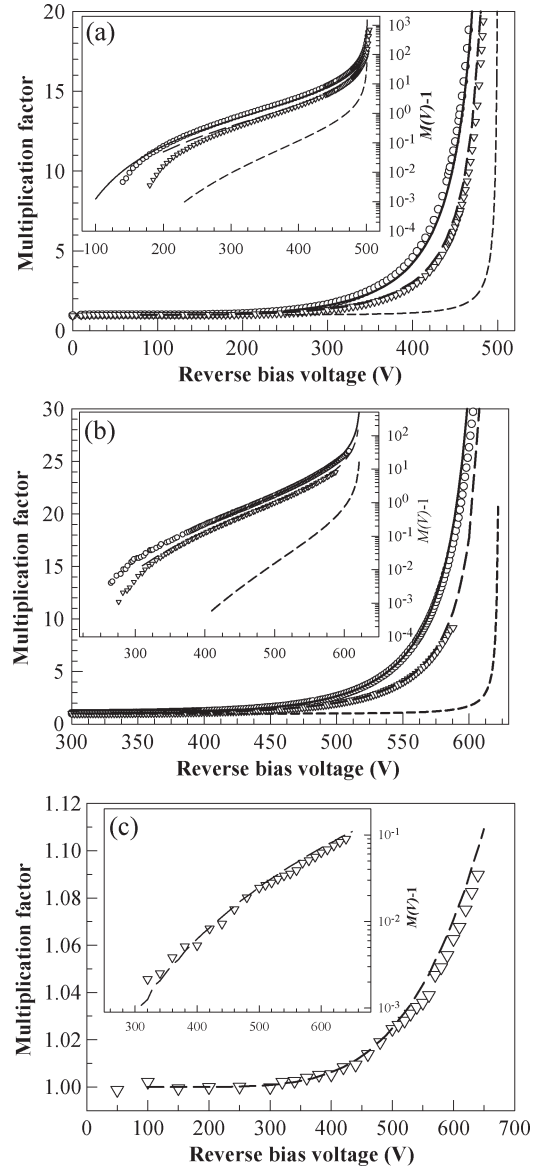


Fig. 3. Multiplication data ($\circ$ for 244 nm, ∇ for 325 nm) of samples (a) A, (b) B, and (c) C. (Solid line) M_h , (short-dashed line) M_e , and (long-dashed line) M_{mix} predicted using the new ionization coefficients are also shown. The insets show the semilogarithmic plot of $[M(V) - 1]$ to reveal the low multiplication values.

of 4H-SiC and, thus, providing reasonably good visible blind performance. It is worth noting that 4H-SiC is an indirect bandgap semiconductor and, thus, does not show the sharp cutoff edge at long wavelength associated with direct bandgap semiconductors such as GaN [15]. The quantum efficiency drops significantly for wavelengths shorter than 250 nm which may due to the direct optical transitions between the valence band and the conduction band near the Γ -point [16]. The surface recombination effect may also contribute to the sharp drop of responsivity for wavelengths less than 250 nm as the penetration depth decreases with decreasing wavelengths of the excitation light.

The measured multiplication characteristics of samples A, B, and C under different carrier injection conditions using 244- and 325-nm excitation light are shown in Fig. 3(a)–(c). The breakdown voltages of samples A and B were estimated to

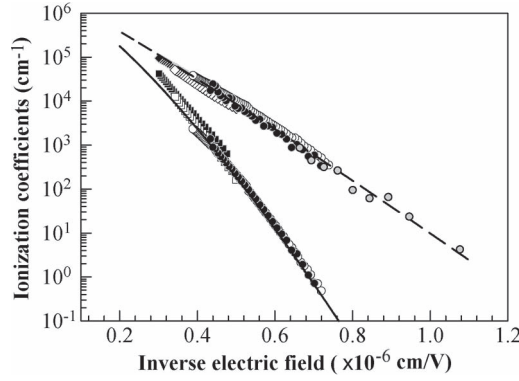


Fig. 4. α and β from samples (○) A, (●) B, and (○) C of this paper along with those for submicrometer devices with avalanche widths of (■) $0.105 \mu\text{m}$ and (□) $0.285 \mu\text{m}$ of [7]. The parameterized (solid line) α and (black-dashed line) β fitted to these data are also shown.

be 504 and 625 V, respectively, by fitting the multiplication profile to Miller's multiplication expression and are in close agreement with the results obtained from the reverse dark I - V measurements. Although the breakdown voltage of sample C is estimated to be greater than 1000 V, measurement under strong avalanching was precluded by the onset of arcing at reverse biases in excess of 650 V. Plots of $[(M(V) - 1)]$ on a semilogarithmic scale are also shown in Fig. 4 to accentuate the low multiplication values at low bias. The multiplication characteristics from shorter wavelength light are consistently higher than those from longer wavelength light for all samples, explicitly indicating the large difference between β and α .

IV. MODELING AND DISCUSSION

Given that the absorption coefficients of 4H-SiC at 325-nm is about 1350 cm^{-1} [17], a significant fraction of the 325-nm light will reach the p^+ -region of all three samples. The multiplication arising from the 325-nm light is mixed (M_{mix}), i.e., both electrons and holes initiate the avalanche process in a manner determined by the incoming light absorption profile. As the illumination wavelength decreases, the light is increasingly absorbed within a short distance of the surface, and thus, more holes are injected from the top n^+ - and n -regions, resulting in primarily hole initiated multiplication characteristics. The multiplication characteristics obtained using 244-nm light with an estimated absorption of 40000 cm^{-1} can be assumed as pure hole initiated multiplication (M_h) in which more than 85% and 90% of the UV light was absorbed in the n^+ - and n -absorption layers of samples A and B, respectively, at the onset of avalanche multiplication. Owing to the large β/α ratio, the small amount of the 244-nm light, which penetrates into the multiplication region, can be neglected and will not affect the accuracy of the determined value of β .

For both samples A and B, the M_{mix} data were found to lie appreciably lower than the calculated M_h using the ionization coefficients and threshold energies in [9]. However, the measured multiplication characteristics using 244-nm excitation were only marginally higher than the calculated M_h . By applying Grant's method [18] to take into account the nonuniform

electric field in each sample, β and α can be deduced from the measured M_h and M_{mix} as described by Ng *et al.* [9].

The ionization coefficients were parameterized by fitting to the high field coefficients of Ng *et al.* [9] and to the lower field coefficients extracted in this work. The field dependence of the ionization coefficients can be expressed as

$$\alpha = 2.78 \times 10^6 \exp \left[- \left(\frac{1.05 \times 10^7}{\xi} \right)^{1.37} \right] \text{ cm}^{-1} \quad (1)$$

$$\beta = 3.51 \times 10^6 \exp \left[- \left(\frac{1.03 \times 10^7}{\xi} \right)^{1.09} \right] \text{ cm}^{-1} \quad (2)$$

where ξ is the electric field in volts per centimeter. The expressions are valid over the electric field range of 0.9–4 MV/cm.

To validate the parameterized ionization coefficients, M_e , M_{mix} (325-nm), and M_h of samples A, B, and C are calculated again using the local model and the electric field profiles obtained from the C - V measurements. Excellent agreement to the measured multiplication characteristics was obtained with the new parameterized ionization coefficients, as shown in Fig. 3. The measured and modeled multiplication characteristics are also plotted as $[(M(V) - 1)]$ on a logarithmic scale to reveal the low multiplication values (see insets of Fig. 3). The small discrepancies between the measured and modeled multiplication characteristics for sample A at low biases may be a result of errors in the normalization of the measured photocurrent from the linear correction.

Fig. 4 shows a summary of the experimentally determined ionization coefficients in bulk structures from the present works and in submicrometer devices from [9], together with the parameterized values given in (1) and (2). It is clear that the experimental results form a consistent set of data covering the electric fields from 0.9 to 4 MV/cm. For sample C, the value of β obtained at low fields is relatively noisy as a result of the low multiplication values. Nevertheless, they follow a similar trend to the extrapolation of β from samples A and B. We note that it is more difficult to accurately determine the exact value of α as the assumptions of the mixed carrier injection profile can generate significant uncertainty in the results. Nevertheless, the measured and calculated M_{mix} 's are in agreement even at low multiplication values, providing further evidence that α has been appropriately determined from this work.

The breakdown voltages of p^+ -i- n^+ structures for a range of avalanche widths W and i-region doping are calculated using a local model and the parameterized expressions in (1) and (2), as shown in Fig. 5. These values are approximately an order of magnitude larger than those for Si. For structures with $W \leq 0.5 \mu\text{m}$, the actual breakdown voltages in SiC are expected to be slightly larger than those predicted by the local model as a result of the dead space effect. Fig. 6 shows a comparison of ionization coefficients obtained from this work with those reported by Konstantinov *et al.* [5], [6], Raghunathan *et al.* [7], [8], Bellotti *et al.* [11], Ng *et al.* [9] and Hatakeyama *et al.* [10]. As shown in Fig. 6, the values of β determined in this work from modeling the experimental results are slightly ($\sim 25\%$) higher than those of Konstantinov *et al.* [5] but lower

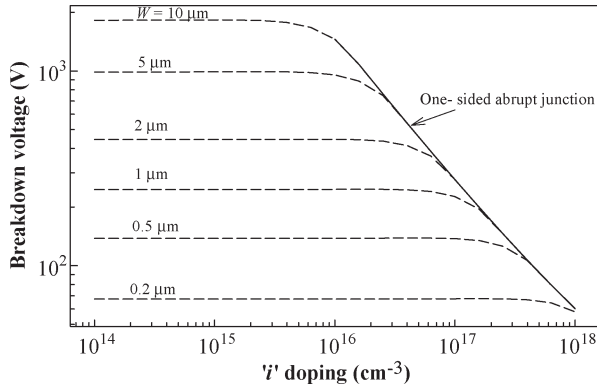


Fig. 5. Calculated breakdown voltage for ideal p^+-i-n^+ diode with different widths and doping for the i -region.

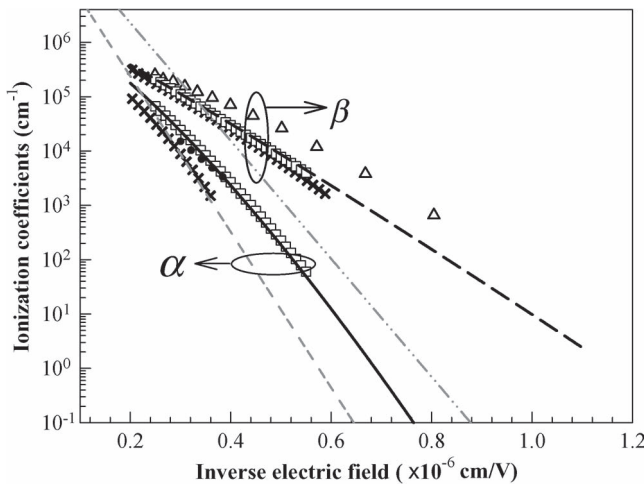


Fig. 6. (Black lines) New parameterized α and β are compared with those by (x) Konstantinov *et al.*, (•) Raghunathan and Baliga, (Δ) Bellotti *et al.*, (gray lines) Hatakeyama *et al.*, and (□) Ng *et al.*

than those of Bellotti *et al.* [11]. Conversely, the deduced α is significantly higher than those of Konstantinov *et al.* [5], particularly at lower fields. It is also worth noting that almost identical α and a slightly lower β ($\sim 15\%$) would be obtained by extrapolating the ionization coefficients in [9] at lower fields, implying that they are in very close agreement with the present results obtained on thick devices.

In Konstantinov's work [5], [6], the multiplication obtained with 325-nm light will result in a significant mixed carrier injection. Furthermore, inaccuracies in the electric field profiles obtained from C - V measurements as mentioned in their work [5], [6] and the insensitivity of the slope of $1/M_h$ to the value of β/α , particularly at $E < 2.7$ MV/cm, will further affect the absolute accuracy of the results obtained. Raghunathan and Baliga [7], [8] reported the value of β by measuring the hole initiated multiplication in 2- μ m-thick p -type Schottky barrier diodes grown on a p^+ substrate with a pulsed electron beam. The assumption that this technique would result in pure hole injection into the high field region may be erroneous because any significant penetration depth of the electron beam would certainly produce mixed injection. This could possibly explain the much lower β value obtained compared with those reported previously and from the work reported here. The value of

electron beam energy and the thickness of the Schottky metal used were not stated by Raghunathan and Baliga [7], [8]; however, for any value in excess of 15 kV, the penetration depth is 2 μ m, and this increases to almost 5 μ m at 25 kV [19]. This would result in primarily minority electrons being injected across the high field and may explain why their data are in much better agreement with the α of Konstantinov *et al.* [5], Ng *et al.* [9], and our recent work. However, in the absence of the actual electron beam conditions used, this is only speculation, and the actual reasons for the discrepancy are unclear. The value of α reported by Hatakeyama *et al.* [10] is similar to that given by Konstantinov *et al.* [5] at lower fields, but β appears to be significantly different with a much stronger dependence on electric field. The work of Hatakeyama *et al.* [10] also suffers from the problem of using 350-nm excitation, which almost certainly gives rise to mixed carrier injection. Furthermore, using a dc photomultiplication measurement with no correction to take in account the small increase in the primary photocurrent at low biases will affect the accuracy of the extracted multiplication data and, thus, the ionization coefficients. By contrast, the photomultiplication measurements performed by Ng *et al.* [9] and in this work used extremely short wavelengths to obtain pure carrier initiated multiplication and a longer wavelength to achieve mixed carrier multiplication. This has enabled both α and β to be determined explicitly within the uncertainties of the absorption coefficients. The good agreement between the ionization coefficients from samples A, B, and C and also the earlier work on submicrometer devices of Ng *et al.* [9] provides further confidence in the accuracy of the data over this wide electric field range.

V. CONCLUSION

Hole and mixed multiplication characteristics were obtained on thick 4H-SiC APDs with nominal avalanche widths between 2.7 and 6 μ m. Ionization coefficients derived from these devices are found to agree well with results obtained on much thinner devices reported by Ng *et al.* [9]. These parameters can be applied to predict the impact ionization mechanism and, therefore, the breakdown voltage in 4H-SiC for electric fields ranging from 0.9 to 4 MV/cm.

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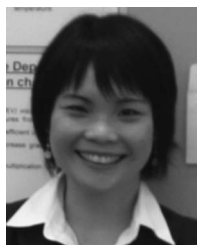
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